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Chapter 08 Stage 8 of 15 Extreme

Transistors and the Front End

Only TSMC, Intel and Samsung make 2 nm-class transistors in volume. TSMC plans A14 for 2028, with more than 20 percent higher logic density than its 2 nm-class N2.

Specimen 08 · Transistors and the Front End

A gate-all-around transistor, the switch in a 2 nm-class chip, drawn at true proportions. The three silicon sheets are each 6 nm thick and 30 nm wide. Each sheet is a few dozen atoms from top to bottom, roughly ten thousand times thinner than a human hair. The gold cage is the gate metal wrapping all four sides of each sheet.

In plain terms

The front end of a chip factory is the part that builds the transistors, the billions of tiny switches in every chip. Each switch works like a valve. Current runs along a narrow strip of silicon called the channel, and a small voltage on a gate above the strip opens or shuts the flow. For decades chipmakers made the strips shorter, so the transistors switched faster and more of them fit on a chip. But once the strip gets short enough, the valve stops sealing. Current leaks through even when the gate is off, and the chip burns power doing nothing. The newest kind of transistor, called gate-all-around, cuts this leak by wrapping the gate around all four sides of the strip. Building them takes a long sequence of steps that have to run in exact order. Only three companies, TSMC, Intel and Samsung, make these switches in volume. In 2024 the United States began requiring a license to share the know-how for building them, except with countries that impose matching controls.

In short

Intel's 18A process shipped gate-all-around transistors together with backside power, a second power network built underneath them[11]. TSMC says its A14 process will give up to 15 percent more speed or up to 30 percent less power than N2[12]. Rapidus, a state-backed newcomer in Japan, said in July 2025 that it would begin 2 nm mass production in 2027[7].

China's SMIC makes 7 nm-class chips without extreme-ultraviolet (EUV) lithography, the most advanced way to print chip patterns[6]. China has

6 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com
7 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc
11 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com
12 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com

announced $43 billion to develop its own EUV [13]. In September 2025 the Center for Strategic and International Studies judged that EUV remains firmly out of reach [13].

Chokepoint card Transistors and the Front End, stage 8 of 15

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

TSMC, Intel and Samsung all make chips with gate-all-around transistors, the newest kind [1].

Price or market size

$17,000

per 300 mm wafer on TSMC's 4NP process, Epoch AI's December 2025 estimate [2]

Who leads	TW	**TSMC**	N2 in volume production since 4Q25, TSMC's first nanosheet node [3]
	US	**Intel**	18A in high-volume production in the United States, as of 2026 [4]
	KR	**Samsung Foundry**	Started initial production of its 3 nm gate-all-around process in June 2022 [5]
	CN	**SMIC**	N+3, a 7 nm-class process made without EUV, SemiAnalysis found in June 2026 [6]
	JP	**Rapidus**	Said in July 2025 it would begin 2 nm mass production in 2027 [7]
Where it is made	TW	**Taiwan**	TSMC N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung
	US	**United States**	Intel 18A at Fab 52, Arizona
	KR	**South Korea**	Samsung Hwaseong and Pyeongtaek
	JP	**Japan**	Rapidus Chitose, Hokkaido
	CN	**China**	SMIC Shanghai and Beijing 300 mm lines

Why substitution is slow

Rapidus, a newcomer, is building on IBM's 2 nm technology under a December 2022 agreement [8]. Since September 2024 the United States has required a license to share the know-how for building these transistors, except with countries that impose matching controls [9].

Where China stands

SMIC makes chips on its N+3 process, which SemiAnalysis measured in June 2026 at 113.4 million transistors per square millimeter, against 107.7 million for TSMC's N6 [6]. It builds its tightest layers in several patterning steps because it cannot buy an extreme-ultraviolet scanner [6].

Where the US stands

In August 2025 Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States [10]. It says its 18A process brought gate-all-around transistors and backside power to market together in 2025 [11].

1 Intel Panther Lake Teardown. SemiAnalysis, 26 September 2026. newsletter.semianalysis.com

2 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

3 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

4 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

5 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

6 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

7 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

8 IBM and Rapidus Form Strategic Partnership to Build Advanced Semiconductor... Rapidus, 13 December 2022. rapidus.inc

9 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

10 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

11 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com

13 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

Nvidia's Blackwell GPU holds 208 billion switches on two dies, each die the largest a lithography machine can print in one shot[14]. Every switch has to turn on fully, let almost no current through when off, and keep doing it for years. Making the switches is called the front end of line. Wiring them together is the back end of line.

01

How it works

Figure 08.1 · Explainer, 23 seconds

Why the gate wraps around the channel



1. **Inside a chip.** An AI chip does all its work with transistors, tiny switches built into the silicon. One Nvidia Blackwell GPU holds 208 billion of them. Making the switches is the front end of the factory, and wiring them together comes after.
2. **On.** Each one is a switch. Seen end on, current flows through a strip of silicon, the channel, and the gate sits on top.
3. **Off, leaking.** Switching off, the gate squeezes the flow shut, but only near itself. In a tiny transistor, current still leaks through the part it cannot reach.
4. **Fin.** Stand the channel on edge as a fin, and the gate covers three sides. Much less gets through.
5. **All around.** Split it into thin sheets and wrap the gate around every side, and it shuts far tighter. That lets the switches keep shrinking, so each new chip can pack in more of them.

Note Simplified. The channel is seen end on, cut across the gate, with current drawn as points of light. Not to scale.

The front end builds the transistors in a fixed order, starting with the channel. In a FinFET, the older design, the fab etches the silicon into thin fins that stand up from the wafer. The gate then goes over the top and both sides of each fin, three sides instead of one.

In a gate-all-around transistor, the fab instead grows alternating layers of silicon and silicon-germanium and cuts the stack into narrow bars[15]. It lays a

14 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

15 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com

placeholder, called a dummy gate, across each bar where the real gate will go[16]. On either side of the placeholder it cuts deep trenches where the source and drain, the two ends of the channel, will go[17]. Through those trenches it etches the silicon-germanium layers back a little and fills the gaps with an insulator, the inner spacer, which keeps the gate apart from the source and drain[15]. Then it grows the source and drain as crystal inside the trenches, a step called epitaxy[17].

Only then does the fab remove the placeholder and etch away the silicon-germanium, leaving thin silicon sheets that hang between the source and drain like shelves[15]. The inner spacer shields the source and drain from that etch[16]. The fab coats every sheet with an insulating film a few atoms thick, so the gate can control the channel without touching it. Gate metal then fills the gaps above, below and beside each sheet, covering all four sides[18]. Last come the contacts, small metal plugs down to the source, the drain and the gate, ready for the wiring layers above.

The steps have to run in exact order, because later layers cover earlier ones. Once gate metal surrounds a sheet, the insulating film can no longer go between the sheet and the metal. Each step is a separate deposition, etch or implant. An implant step, called ion implantation, fires atoms of boron or phosphorus into chosen patches of silicon so they conduct the way the design needs.

Current leaks in two main ways, straight through the gate insulator and along the channel from end to end while the gate is off. The second grows as the channel gets shorter, and wrapping the gate around the channel reduces it. Three fixes deal with the two leaks:

- **Better insulator.** A thicker hafnium-based film, paired with a metal gate, lets fewer electrons slip straight through yet controls the channel as tightly as before. Intel replaced the silicon dioxide insulator with it at 45 nm in 2007[19]. It said the new film cut leakage through the insulator more than tenfold[19]. With the metal gate, it could either raise the current a switch carries when on by more than 20 percent or cut leakage along the channel more than fivefold[19].
- **More gate.** Fins and nanosheets wrap gate metal around more sides of the channel.
- **Thinner channel.** The fab thins the silicon until no current can flow where the gate cannot reach it.

15 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com

16 Development of SiGe Indentation Process Control for Gate-All-Around FET... ArXiv, 12 January 2022. arxiv.org

17 Applied Materials Unveils Next-Gen Chipmaking Products to Supercharge AI... Applied Materials, 7 October 2025. ir.appliedmaterials.com

18 Applied Materials Technologies Enable 2D Scaling with EUV and 3D... Applied Materials, 21 April 2022. ir.appliedmaterials.com

19 Intel's Transistor Technology Breakthrough Represents Biggest Change to... Intel, 27 January 2007. intel.com

02

Variants and trade-offs

Planar and FinFET

A planar transistor lays the gate on one side of a flat channel. As the channel shortens, the gate loses control of it. A fin stands the channel on edge so the gate covers three sides. But each fin adds a fixed amount of current, so a designer who needs more current has to add a whole fin.

Gate-all-around nanosheets

A nanosheet transistor lays the channel flat again, stacks two to four thin sheets and wraps gate metal around each one. The sheets can be drawn at different widths, so current can be set in finer steps. On N2, TSMC offers NanoFlex, which lets designers mix short and tall standard cells, the basic building blocks of chip design[20]. Short cells save area and power, and tall cells give the most performance[20].

TSMC's N2, its first nanosheet node, entered volume production in the fourth quarter of 2025[3]. TSMC rates the follow-on N2P at 18 percent more speed at the same power and 36 percent less power at the same speed than its earlier N3E[21]. It gives N2P 1.2 times the logic density of N3E and 1.15 times its chip density, but its technology page does not define either measure[21].

Intel and Samsung also make gate-all-around transistors. Intel's version, called RibbonFET, is built into its 18A process, which Intel says is in high-volume production in the United States[4]. An improved version, 18A-P, adds another 9 percent performance at the same power[11]. Samsung got there first, starting initial production of its 3 nm gate-all-around process in June 2022[5]. But SemiAnalysis reported in February 2023 that, because of yield problems, high-volume chips on Samsung's 3 nm gate-all-around technology were expected only in 2024[22].

Those numbers are the foundries' own claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure that can be compared with a rival's.

3 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

4 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

5 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

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20 TSMC Celebrates 30th North America Technology Symposium with Innovations... TSMC, 24 April 2024. pr.tsmc.com

21 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com

22 The Future of the Transistor. SemiAnalysis, 21 February 2023. newsletter.semianalysis.com

Figure 08.2

N2P gains 15 percent chip density and A16 up to 10 percent

TSMC's published chip density gains, each over the node TSMC names, with A16 at its stated maximum



Source TSMC advanced technologies, HPC platform [21]

Backside power delivery

In a conventional chip, power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. With backside power, the fab grinds the wafer thin and builds a second network for power underneath it.

Intel's version, PowerVia, is part of 18A, and Intel calls it an industry first[4]. It cut routed area, the space a block of circuits takes once its wiring is laid out, by 11 percent and dynamic voltage droop tenfold[11]. Droop is the sag in supply voltage when a block of logic switches on at once. TSMC's Super Power Rail connects the backside network directly to each transistor's source and drain, the two ends of its channel[23]. Against N2P, TSMC rates A16, which uses Super Power Rail, at 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density[21].

TSMC plans to start A14 production in 2028, with more than 20 percent higher logic density than N2[12]. It said in April 2026 that two more processes, A13 and A12, are scheduled to enter production in 2029[24]. A13 is a direct shrink of A14 with 6 percent area savings, and A12 adds Super Power Rail to the A14 platform[24].

4 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

11 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com

12 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com

21 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com

23 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com

24 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

Figure 08.3

TSMC plans four logic nodes beyond N2 for 2028 and 2029

TSMC's logic roadmap beyond N2, from its own announcements

- 2028
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028
  A14, the base node for A13 and A12
- 2029
  A13, a direct shrink of A14 for 6% area savings
- 2029
  A12, the A14 platform with Super Power Rail backside power

Source TSMC, 2026 North America Technology Symposium [24]

CFET and what comes after

A basic logic circuit needs two transistors that switch on opposite signals, and in production chips the two sit side by side. Complementary field-effect transistors, or CFETs, stack the two on top of each other[25]. The research institute imec sees them as an attractive design for logic nodes beyond 1 nm[25].

03

The back end of line

Wiring has become as hard to make as the transistor. Copper seeps into the insulator around it, so every wire needs a barrier, a liner and a cap[26]. As the most critical wiring levels drop below a 20 nm pitch, those extra layers take a large share of the wire's width[26]. Pitch is the distance from the center of one wire to the center of the next. Two metals are moving into those narrow levels.

- **Ruthenium** shows promise for use without a barrier, which would keep the wire's resistance low[26]. imec has demonstrated ruthenium lines at a 16 nm pitch[27].
- **Molybdenum** conducts worse than copper in bulk, with a resistivity of 5.3 microhm-cm against copper's 1.68[28]. But in a thin film it has up to 30 percent lower resistivity than tungsten, the metal it is replacing in contacts[28]. It needs no barrier either, so chipmakers are putting it into some contacts and local wiring layers[28].

24 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
25 CFET (complementary FET). Imec, 16 June 2022. imec-int.com
26 Semi-damascene metallization. Imec, 31 October 2024. imec-int.com
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28 Molybdenum's Role in Ultra-Fast Computing: The Metal Behind the Speed. Entegris, 18 May 2026. blog.entegris.com

04

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme-ultraviolet (EUV) lithography. China's SMIC builds 7 nm-class nodes without it[6].

SemiAnalysis reported in September 2026 that Intel's Panther Lake, Samsung's Exynos 2600 and Apple's A20 Pro, which is made on TSMC's N2, all use gate-all-around transistors[1]. In August 2025 Intel called itself the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[10].

In July 2025 Rapidus, a state-backed newcomer in Japan, said it had started making 2 nm prototypes at its plant in Chitose, Hokkaido, and would begin mass production in 2027[7]. Its process builds on IBM's 2 nm technology, which the two firms agreed in December 2022 to develop further for production at Rapidus's plant in Japan[8]. By June 2026 Rapidus had raised 424.95 billion yen in equity[29]. The most recent 150 billion yen came from the Information-Technology Promotion Agency, a state agency under Japan's industry ministry[29]. It has also received government research subsidies since fiscal 2022[29].

A node's name, such as 7 nm, does not measure any feature on the chip. Applied Materials, a toolmaker, said in 2022 that chipmakers raise density in two ways, by printing smaller features and by laying out each logic cell more tightly[18]. SMIC's N+3, its third-generation 7 nm process, sets its narrowest wires 32.5 nm apart, tighter than the 36 nm in Intel's Panther Lake chips on 18A[6]. But the 18A process itself allows wires 32 nm apart[6]. SemiAnalysis called the comparison an incomplete, cherry-picked metric[6]. N+3 packs 113.4 million transistors per square millimeter, against 107.7 for TSMC's N6[6].

AI accelerators have usually trailed the leading edge. A die as large as theirs is more likely to hold a defect[30]. So the share of dies that work, called the yield, gets high enough only once a process has run long enough to cut its defect rate.

Nvidia's Blackwell is made on an older TSMC process than AMD's MI455X. Blackwell uses a custom TSMC 4NP process[14]. Epoch AI estimated in December 2025 that a 300 mm wafer on that process costs $17,000[2]. AMD's MI455X, launched on 23 July 2026, is made on TSMC's 2 nm and 3 nm processes[31]. It spreads its compute across eight separate dies[32].

1 Intel Panther Lake Teardown. SemiAnalysis, 26 September 2026. newsletter.semianalysis.com

2 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

6 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

7 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

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14 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

18 Applied Materials Technologies Enable 2D Scaling with EUV and 3D... Applied Materials, 21 April 2022. ir.appliedmaterials.com

29 Rapidus Completes 150 Billion Yen Funding Round from Japan Government. Rapidus, 5 June 2026. rapidus.inc

30 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

31 AMD Instinct™ MI455X GPUs. AMD, accessed 9 September 2026. amd.com

32 AMD Instinct MI455X GPU. AMD, 20 July 2026. amd.com

SemiAnalysis reported in 2024 that the SRAM cell, which holds one bit in the fast memory on a logic die, had barely shrunk since the 5 nm node[23]. TSMC said in 2025 that its 2 nm SRAM uses a 0.021 square-micrometer cell and gets 1.1 times the density of its previous node by tuning design and process together[33]. See Memory and HBM.

05

The chokepoint

TSMC, Intel and Samsung do not sell their leading-edge process flows, which set the steps, their order and the tools for each. In September 2024 the United States began requiring a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls[9].

China has money for chipmaking but cannot buy EUV scanners. It launched its third National Integrated Circuit Fund in May 2024 with 344 billion yuan, about $47 billion[34]. China's 7 nm chips have been made on deep-ultraviolet machines, a generation behind EUV[13]. China had stockpiled those machines from ASML, the Dutch scanner maker, before export restrictions took effect[13]. SMIC gets fine features out of them by building its tightest layers in several patterning steps[6].

In July 2026 Reuters reported that a Shanghai state firm had begun making home-grown immersion deep-ultraviolet scanners[35]. Citing The Information, it said the firm planned about five that year and roughly 20 in 2027[35]. Reuters's own source said the firm's machine still needed testing and was far from matching ASML's models[35].

Domestic EUV is further off. China has announced $43 billion to develop it[13]. In September 2025 the Center for Strategic and International Studies (CSIS) judged that EUV remains firmly out of reach despite massive state investment[13].

See Lithography and Who Controls the Chain.

Four questions to answer from memory

1. Why do the newest transistors wrap the gate around the channel?
   `see How it works`
2. Which three companies make gate-all-around transistors in volume?
   `see Who makes it`
3. Why is it hard for a new company to make leading-edge transistors?
   `see The chokepoint`

6 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. `newsletter.semianalysis.com`

9 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. `federalregister.gov`

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35 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. `reuters.com`

4. How does China's SMIC make advanced chips without an EUV machine?
   see The chokepoint

Four things to remember

- The newest transistors wrap the gate around all four sides of the channel to cut the current that leaks when the switch is off.
- Only TSMC, Intel and Samsung make them in volume.
- TSMC, Intel and Samsung do not sell their leading-edge process flows, so a newcomer has to develop its own or build on another firm's, as Rapidus is doing with IBM's.
- China's SMIC makes 7 nm-class chips without EUV by building its tightest layers in several patterning steps.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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